

Chemistry

Unit 4

Practice Exam Answers

Section 1: Multiple choice

(32 marks)

Question 1

- C** A is incorrect because alkanes do not exhibit *cis-trans* isomerism. B should be 3-methylpentan-2-ol. C is correct; numbers are unnecessary as the structure is unambiguous. D is incorrect because a ketone does not have its functional group at the end of the chain; the name should be butanal.

Question 2

- D** The longest carbon chain has five carbon atoms so it is a pentane derivative. The numbering of carbons in the parent chain is from the end, which gives the side chains and functional group the smallest number. So the -OH group is on carbon number 1 and the two methyl groups are on carbon number 3.

Question 3

- A** Propanone is not oxidised by acidified dichromate, so no reaction will occur. The mixture will have the colour of the unreacted dichromate (i.e. orange).

Question 4

- A** Only dispersion forces operate between the alkane molecules. As butane is larger (and contains more electrons), its boiling point is higher than that of propane. Butanone has a polar C=O group and so dipole-dipole interactions operate between its molecules. So the boiling point of butanone is higher than that of similar-sized alkanes.

Butan-1-ol has the highly polar -O-H group, so there is some hydrogen bonding between molecules. Therefore, butan-1-ol has the highest boiling point.

Question 5

- D** Ethanoic acid is a weak acid and forms a solution with a pH of less than 7. Ethanol is not hydrolysed in water, so it forms a solution with a pH of 7. Like ammonia, ethanamine is a weak base and forms a solution with a pH greater than 7.

Question 6

- C** The amino acids that condense to form a protein are α -amino acids; that is, amino acids in which the -COOH and -NH_2 groups are on the same C atom. In C, they are on different carbon atoms.

Question 7

- B** Amino acids act as acids–bases in aqueous solutions. At a high pH, with an excess of OH[−] ions, the amino acid will act as an acid, donating a proton from the COOH group.

Question 8

- C** The secondary structure is maintained by hydrogen bonds between amide and carbonyl functional groups. In a pleated sheet structure, these are between groups in adjacent polypeptide chains.

Question 9

- C** Ethene is unsaturated and undergoes an addition reaction with water.
 $\text{CH}_2=\text{CH}_2 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$, so the molecular formula is C₂H₆O.

Question 10

- A** The reaction is endothermic, so an increase in temperature will shift the equilibrium to the right, thus producing more product. There are 2 moles of reactants in the equation but 4 moles of products. Therefore, an increase in pressure will favour the reverse reaction and the amount of H₂(g) will decrease. A catalyst only affects the reaction rate, not the product yield.

Question 11

- D** The possible tripeptides are:
Gly–Gly–Ala Ala–Gly–Gly Gly–Ala–Gly Ala–Ala–Gly
Ala–Gly–Ala Gly–Ala–Ala Gly–Gly–Gly Ala–Ala–Ala

Question 12

- B** Primary alcohols are oxidised to acids. When acids react with alcohols to form an ester, water is eliminated and so the reaction is a condensation reaction. The primary alcohol drawn is propan-1-ol. This would be oxidised to propanoic acid. Therefore, the ester would be propyl propanoate.

Question 13

- A** The overall percentage yield is calculated by multiplying the percentage yields of each step.
Therefore, the overall percentage yield is $\frac{18}{100} \times \frac{70}{100} \times \frac{10}{100} \times 100 = 1.3\%$.

Question 14

- A** The rate of reaction increases, and continues to increase, as both pressure and temperature increase.

Question 15

- D** Increasing pressure has no effect on the catalyst nor on the stability of SO₃(g). As there are 3 moles of gas on the left of the equation and only 2 moles on the right, an increase in temperature would favour the formation of SO₃(g). By carefully selecting the proportion of reactants, a suitable catalyst and temperature, a commercially viable conversion to SO₃ is achieved without the extra costs of building equipment to withstand higher pressures.

Question 16

- B** The simplest alkene is CH₂=CH₂ (ethene). The simplest ketone is CH₃C(=O)CH₃ (propanone). The simplest ester is HCOOCH₃ (methyl methanoate). The simplest aldehyde is HCHO (methanal). Only ketones must have a minimum of three carbon atoms.

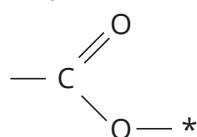
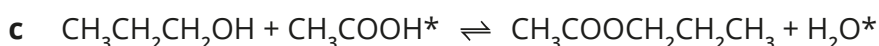
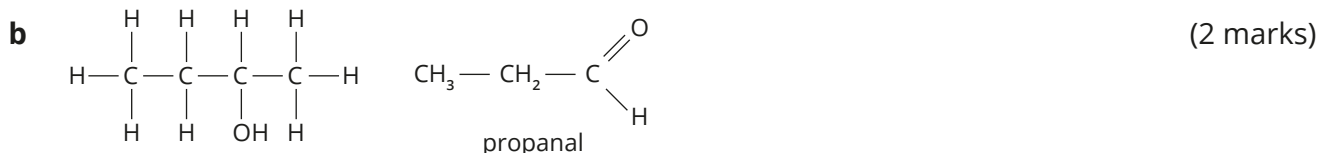
Section 2: Short answer

(50 marks)

(Note: In multi-mark questions each allocated mark is indicated as * and total marks are indicated in parentheses.)

Question 17

a butan-2-one*, *cis*-pent-2-ene*, ethanamide* (3 marks)



propyl ethanoate*

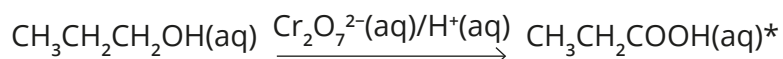
(4 marks)

Question 18

a Test 1

Add a few drops of orange acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$ to both*.

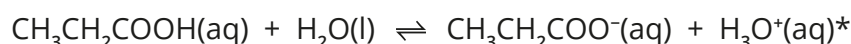
The solution will turn green with the propan-1-ol*.



Test 2

Add some litmus (or other suitable indicator)*.

The litmus will turn pink in the acid solution*.



OR

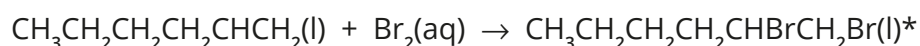
Other alternatives would include adding carbonate powder or a small piece of Mg and noting the evolution of gas or using a pH meter.

(6 marks)

b Test 1

Add some brown bromine water to both*.

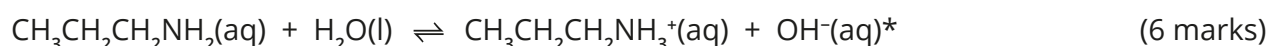
With hex-1-ene the brown colour will disappear*.



Test 2

Add some litmus (or other suitable indicator)*.

The litmus will turn blue in propan-1-amine*.



Question 19

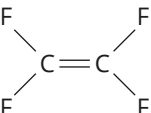
a alkene*; alcohol (or hydroxyl or hydroxy)* (2 marks)

b Vitamin C has a much higher proportion of polar $-\text{OH}$ groups*, which can form hydrogen bonds with water and allow vitamin C to be water soluble*. (2 marks)

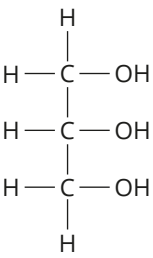
Question 20

- a The carboxylate group (COO^-) has been replaced by a sulfonate group (SO_3^-). (1 mark)
- b anionic (1 mark)
- c non-polar (1 mark)
- $\text{CH}_3(\text{CH}_2)_{10}\text{CH}_2$ —— SO_3Na
- d Hard water contains calcium ions*. These form a precipitate (scum) with soap but not with the above detergent*. (2 marks)

Question 21

- a  (1 mark)
- b $M(\text{C}_2\text{F}_4) = 100 \text{ g mol}^{-1}$
The polymer is an addition polymer of formula $-(\text{C}_2\text{H}_4)_n-$. So the average number of C_2H_4 units in the polymer is as follows.
$$M(\text{polymer})/M(\text{C}_2\text{F}_4) = \frac{1\,500\,000 \text{ g mol}^{-1}}{100 \text{ g mol}^{-1}} = 15\,000 \text{ monomer units}^*$$
 (2 marks)
- c For example, heat resistant/coating on cookware, low friction/ball bearings etc. (1 mark for property and 1 mark for use related to that property) (2 marks)

Question 22

- a CH_3OH (1 mark)
- b i 
glycerol (2 marks)
- ii Glycerol is highly polar so will dissolve in an aqueous layer*, whereas biodiesel, being non-polar, is water insoluble. It forms a layer over the aqueous solutions and this can be decanted off*. (2 marks)
- c NaOH (or KOH) (1 mark)
- d i enzyme or protein (1 mark)
- ii An advantage is that large-scale use of a corrosive substance such as NaOH is avoided*. A disadvantage is that the enzyme can become deactivated*. (2 marks)

Question 23

- a $n(\text{Fe}_2\text{O}_3) = \frac{5.00 \times 10^8}{159.7} = 3.13 \times 10^6$
 $n(\text{CO}) = \frac{3.50 \times 10^8}{28.01} = 1.25 \times 10^7$
Each mole of Fe_2O_3 requires 3 moles of CO
so $n(\text{CO})$ required = $3 \times 3.13 \times 10^6 = 9.39 \times 10^6$
As there is more CO present than the amount required, the Fe_2O_3 is the limiting agent*. (3 marks)
- b theoretical $n(\text{Fe})$ produced = $2 \times 3.13 \times 10^6 = 6.26 \times 10^6$
theoretical $m(\text{Fe})$ produced = $6.26 \times 10^6 \times 55.85 = 3.50 \times 10^8 \text{ g} = 350 \text{ tonnes}^*$
 $\% \text{ yield} = \frac{320}{350 \times 100} = 91.4 \%$ (3 sig. figs)* (3 marks)

Section 3: Extended answer

(38 marks)

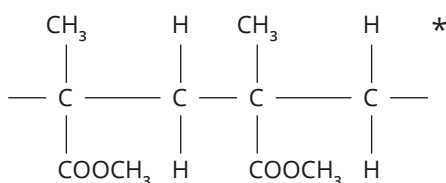
(Note: In multi-mark questions each allocated mark is indicated as * and total marks are indicated in parentheses.)

Question 24

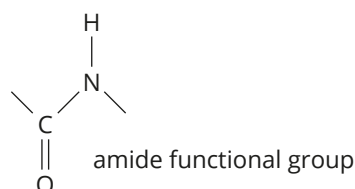
- a i** Condensation polymers are formed between monomers that each have two reactive functional groups. A small molecule, often water, is eliminated in the formation of the polymer*. Monomers that undergo addition polymerisation each have at least one double bond*. (2 marks)

- ii** Relatively strong hydrogen bonds form between the many C=O groups of one chain and the many N—H groups of an adjacent chain, thus forming a strong polymer. (1 marks)

- iii** addition polymerisation* (2 marks)



- b** (1 mark)



- c i** nitrogen (1 mark)

$$\begin{aligned} \text{ii} \quad n(\text{C}):n(\text{H}):n(\text{O}):n(\text{N}) &= \frac{34.3}{12.01} : \frac{6.7}{1.008} : \frac{45.7}{16.00} : \frac{[100 - (34.4 + 6.67 + 45.7)]}{14.01} * \\ &= 2.85:6.65:2.86:0.93 \\ &= 3:7:3:1 \end{aligned}$$

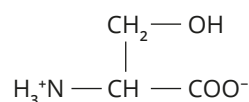
so the empirical formula is $\text{C}_3\text{H}_7\text{O}_3\text{N}^*$

Since 0.250 mol have a mass of 26.26 g, molar mass of amino acid
 $= m/n = 26.26/0.250 = 105.04 \text{ g mol}^{-1}$ *

formula mass of $\text{C}_3\text{H}_7\text{O}_3\text{N} = 105.05 \text{ g mol}^{-1}$

so molecular formula = $\text{C}_3\text{H}_7\text{O}_3\text{N}^*$ (4 marks)

- iii** As all α -amino acids contain a $\text{H}_2\text{NCHCOOH}$ backbone, the atoms left to form the side group are CH_3O so the amino acid is serine. In solution at pH 7, this is most likely to be in the form below.



(1 mark for identifying serine and 1 mark for showing it as a zwitterion)

Amino acids	Interaction (covalent, ionic or hydrogen bonds, dipole-dipole interactions or dispersion forces)
threonine and lysine	hydrogen bonds*
cysteine and cysteine	covalent bond*
valine and leucine	dispersion forces*
aspartic acid and lysine	ionic interactions*

(4 marks)

- ii At high temperatures, the kinetic energy of the molecule is sufficient to disrupt the bonding between the side groups of amino acid residues*, so the tertiary structure is lost*.

High pH affects the charges of amino acid residues with acidic or basic side chains*, thus reducing the interactions between those amino acids (with the loss of some of the tertiary protein structure)*. (4 marks)

Question 25

- a i $K = \frac{[N_2][H_2]^3}{[NH_3]^2}$ (1 mark)
- ii The reaction is exothermic, so an increase in temperature favours the reverse reaction*. As there is a smaller concentration of reactants and larger concentration of products, the value of K will decrease as the temperature increases*. (2 marks)
- b i As the reaction is exothermic, increased yield is favoured by low temperatures but the reaction rate is too slow if the temperature is too low*. The temperature chosen is a compromise between an acceptable yield and rate*. (2 marks)
- ii At higher pressures, reactants particles are closer together, increasing the likelihood of collisions*, and so the reaction rate increases*. As there are 4 moles of reactant molecules but only 2 moles of product molecules*, an increase in pressures pushes the equilibrium to favour the forward reaction thus increasing the yield of the product*. (4 marks)
- iii A catalyst provides a reaction pathway of lower activation energy*, so a greater proportion of reactant molecules will have energy greater than or equal to the activation energy required to react*. A porous catalyst provides a large surface area to maximise contact with reactants*. (3 marks)
- c Ammonia (unlike hydrogen and nitrogen) is highly polar* and so it forms hydrogen bonds between its molecules, allowing it to liquefy more easily*. (2 marks)
- d $n(NH_3) \text{ required} = \frac{800\,000}{17.034} = 4.70 \times 10^4$
 so $n(H_2)$ theoretically required $= \frac{3}{2 \times 4.70 \times 10^4} = 7.05 \times 10^4$ *
 but as yield is 98%
 so $n(H_2)$ actually required $= \frac{100}{98 \times 7.05 \times 10^4} = 7.19 \times 10^4$ *
 so $V(H_2) = \frac{nRT}{P} = \frac{7.19 \times 10^4 \times 8.314 \times 298.15}{100} = 1.78 \times 10^6 \text{ L}^*$ (3 marks)

End of answers